template for a protein or RNA molecule (in case of tRNA or rRNA genes). Each gene contain a distinct sequence of nucleotides which determines the aminoacid sequence or nucleotide sequence a cell might synthesize. If a gene i.e, a segment of DNA can act as a template to synthesize an RNA transcript by the process of translation, that gene is considered to be expressed.

6.3.2 Overview of transcription

1. It is a process where a DNA sequence is read by an enzyme called RNA polymerase and a complementary, antiparallel RNA strand is synthesized.
2. Apart from RNA polymerase many proteins called transcription factors control the rate of transcription by binding to specific DNA sequence.
3. Transcription between eukaryotes and prokaryotes will vary slightly.
4. Some viruses (eg. HIV) have the ability to transcribe RNA into DNA. HIV has an RNA genome that is reverse transcribed into DNA.
5. RNA polymerase, together with one or more general transcription factors, binds to promoter DNA.
6. RNA polymerase creates a transcription bubble, which separates the two strands of the DNA helix. This is done by breaking the hydrogen bonds between complementary DNA nucleotides.

There are three stages in transcription which are initiation, elongation and termination.

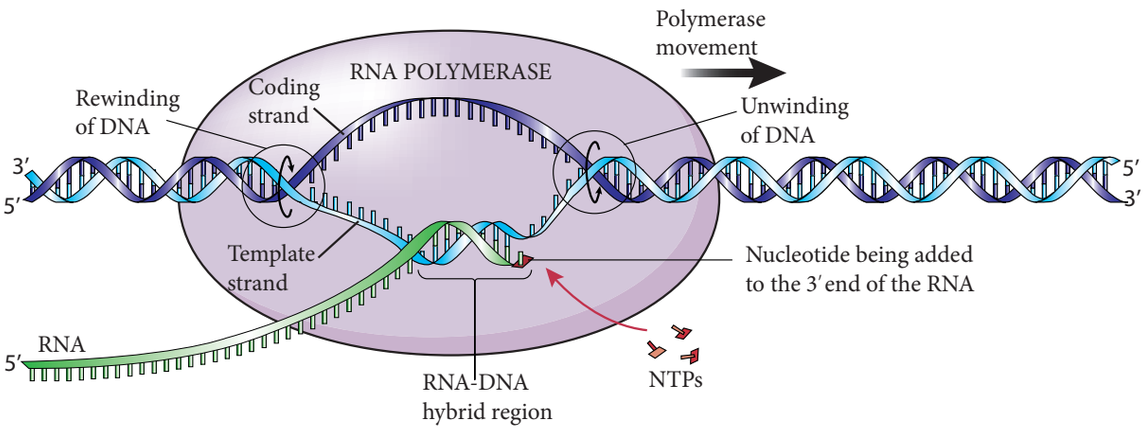


Figure 6. Representation of the transcription process

6.3.2.1 Initiation of transcription

1. RNA polymerase, together with one or more general transcription factors, binds to promoter DNA.
2. RNA polymerase, along with transcription factor sigma, binds to a sequence of DNA called the promoter, found near the beginning of a gene.

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